AttentiQ: Does Attention Survive Quantization?

Trustworthy Deepfake Detection at the Edge

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Abstract. Deepfakes threaten digital trust, and detectors are increasingly needed on resource-constrained devices. Attention modules such as CBAM improve visual feature selection, while INT8 quantization reduces cost; yet CBAM relies on sigmoid gating and element-wise multiplication, operations that are sensitive to low-precision arithmetic. AttentiQ asks whether the benefit of attention *survives* quantization. We compare a ResNet-18 baseline, an SE-attention control, a CBAM-enhanced ResNet-18, its post-training INT8 version, and a quantization-friendly CBAM variant trained with quantization-aware training. Beyond accuracy and cost, we measure *attention fidelity*—how much quantization shifts where the model looks—and cross-dataset generalization from FaceForensics++ to Celeb-DF. Latency is measured on a real edge device. All targets are exploratory; no results are reported at the proposal stage.

1 Motivation

Face-manipulation tools now produce convincing fake images and videos, enabling misinformation and impersonation. Detection must increasingly run close to the user—on phones, cameras, and moderation devices—where memory and compute are limited. Two common tools address this: *attention* to focus the detector on subtle manipulation artifacts, and *quantization* to shrink and accelerate the model. These tools are usually studied separately. Their interaction is the focus of this project (Figure 1): if quantization distorts the attention maps, the accuracy gained from attention may be silently lost on the edge.

2 Related Work and Research Gap

FaceForensics++ [10] established the standard benchmark, and Celeb-DF [9] and DFDC [2] exposed the poor cross-dataset generalization of detectors. MesoNet [1] showed that compact networks can detect deepfakes, and multi-attentional detection [13] showed that attention helps locate subtle artifacts. Residual networks [4], squeeze-and-excitation (SE) channel attention [6], and CBAM channel-plus-spatial attention [12] provide the backbone and attention modules. Integer quantization [7, 8] enables efficient inference, MobileNetV3 [5] replaced sigmoid gating with hard-sigmoid because sigmoid is costly to compute and hard to keep accurate in fixed-point arithmetic.

Gap. Prior work reports accuracy *or* efficiency, but rarely asks whether attention mechanisms remain faithful after quantization. For deepfake detection—where decisions depend on small, localized artifacts—a shift in spatial attention may matter more than a small change in overall accuracy. This project measures that interaction directly.

3 Research Questions

RQ1 Does CBAM improve deepfake detection over ResNet-18, beyond what channel-only attention (SE) provides?

RQ2 Does INT8 quantization reduce size and edge latency while preserving the attention gain?

RQ3 Does quantization change *where* CBAM attends, and can a quantization-friendly CBAM reduce this change?

RQ4 Does attention reduce the performance drop on an unseen dataset?

4 Proposed Approach

The models are summarized in Table 1.

Table 1: Models compared. B1 is a lightweight reference; M3 is a stretch goal.

ID	Architecture	Purpose
B1	MesoNet (Meso-4)	Compact detector reference
M0	ResNet-18 (FP32)	Baseline
C1	SE-ResNet-18 (FP32)	Control: channel-only attention
M1	ResNet-18 + CBAM (FP32)	Attention model
M2	M1, post-training static INT8	Efficiency (PTQ)
M3	ResNet-18 + QF-CBAM, QAT INT8	Quantization-friendly attention

Part 1 — Attention. CBAM is inserted after each residual stage of an ImageNet-pretrained ResNet-18. C1 places SE blocks at the same positions, so that the comparison M1 vs. C1 isolates the contribution of *spatial* attention, and M0 vs. C1/M1 isolates attention from simple capacity. Parameter counts and MACs are reported for every model. A small placement ablation (CBAM after stages 3–4 only vs. all stages) is run if time allows.

Part 2 — Quantization. M2 applies post-training static INT8 quantization to M1, using the `torchao` PT2E flow (`prepare_pt2e`/`convert_pt2e`; the older `torch.ao.quantization` FX flow is deprecated) or ONNX Runtime static quantization for edge export, calibrated on a held-out portion of the training set. Operator support is checked layer by layer; any operation left in floating point (e.g., sigmoid gating) is reported explicitly.

Part 3 — Quantization-friendly attention (stretch).

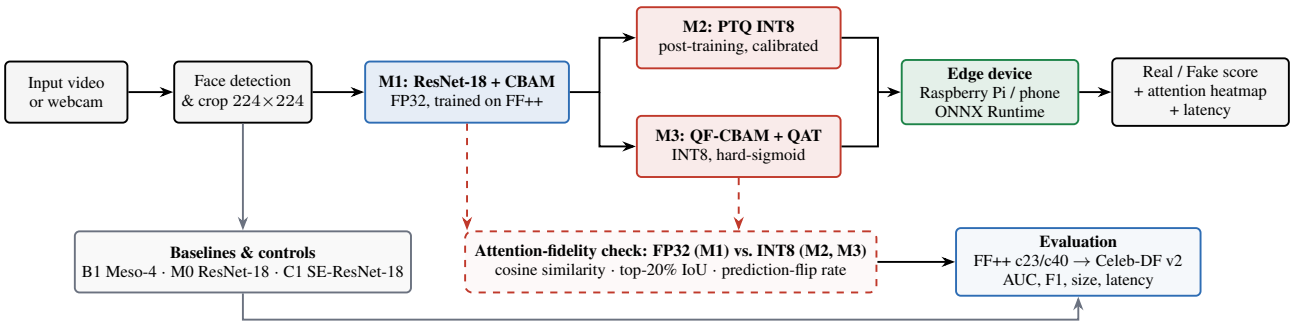


Figure 1: AttentiQ pipeline. A CBAM-enhanced ResNet-18 (M1) is quantized two ways—post-training (M2) and with quantization-aware training of a quantization-friendly CBAM (M3)—and deployed on an edge device. The attention-fidelity check measures whether INT8 changes *where* the detector looks, not only how accurate it is.

M3 replaces CBAM's sigmoid with hard-sigmoid and its pooling-then-multiply path with quantization-supported operators (QF-CBAM), and is trained with quantization-aware training (QAT, via `prepare_qat_pt2e`). This tests whether a small design change restores what PTQ loses (Figure 2).

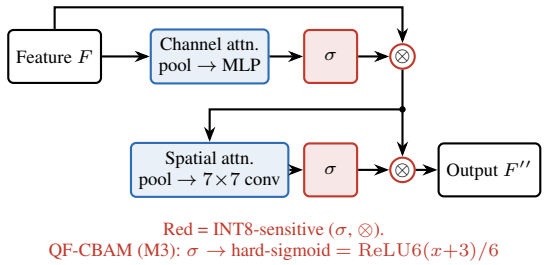


Figure 2: CBAM block and the quantization-sensitive operations (red) that M3 modifies.

Part 4 — Attention fidelity. For each test image we extract CBAM's spatial attention maps (and Grad-CAM [11] maps for all models) and compute, between FP32 M1 and its INT8 versions: mean cosine similarity, IoU of the top-20% attended region, and the fraction of predictions that flip. We also check whether attention concentrates on face regions (eyes, mouth, blending boundary) rather than background.

5 Hypotheses

All targets are exploratory expectations, not established findings. Effects are judged on the mean over three seeds and must exceed the seed standard deviation.

H1 — Attention. M1 improves frame-level AUC on the FF++ (c23) test set by $\geq$2 percentage points over M0, and by more than C1 does.

H2 — Efficiency. M2 reduces model size by $\geq$50% and edge-device latency by $\geq$20% relative to M1, while retaining $\geq$50% of M1's AUC gain over M0.

H3 — Attention fidelity. PTQ (M2) measurably alters CBAM spatial attention (mean cosine similarity to M1 below 0.90), and M3 preserves attention more closely than M2.

H4 — Generalization. The AUC drop from FF++ to Celeb-DF is smaller for M1 than for M0.

6 Datasets and Experimental Design

Data. FaceForensics++ [10] (1,000 real videos; Deepfakes, Face2Face, FaceSwap and NeuralTextures manipulations) is used for training, validation and in-domain testing, at c23 (primary) and c40 (heavy compression, simulating social-media video). Celeb-DF v2 [9] is the external test set; the DFDC preview set [3] is the alternative if access is delayed. The full DFDC ($\approx$470 GB) is out of scope.

Preprocessing. Faces are detected and cropped (RetinaFace or MTCNN) with a fixed margin and resized to 224×224. A fixed number of frames is sampled per video (e.g., 8 per real video and 2 per manipulated video, to balance classes), yielding roughly 4,000–6,000 training images.

Splits and leakage control. The official FF++ video-level split (720/140/140) is used, so a source video and all its manipulated versions always fall in the same partition. Validation guides model selection; quantization calibration uses only training data; the test sets are used once, for final reporting.

Training protocol. All FP32 models share resolution, augmentation, optimizer, schedule, epoch budget and early-stopping rule. Each model is trained with three random seeds (recorded).

Metrics. *Detection:* AUC (primary), F1, accuracy, precision, recall; at frame level and video level (mean of frame scores per video); reported per manipulation type. *Efficiency:* model size on disk, parameters, MACs, peak memory, and latency. *Fidelity:* attention cosine similarity, top-20% IoU, prediction-flip rate.

Latency protocol. Latency is measured on a real edge device (Raspberry Pi 4/5, or an Android phone via ONNX Runtime/TFLite) and on a laptop CPU, with batch size 1, single thread and multi-thread settings, 50 warm-up runs and the median of 500 timed runs. Preprocessing is excluded and reported separately.

7 Scope and Feasibility

Core (required): B1, M0, C1, M1, M2; H1, H2, H4; attention-fidelity analysis of M2; three seeds; latency on at least one edge device or, failing that, a laptop CPU with the limitation stated.

Stretch: M3 with QAT (H3, second part); c40 robustness;

placement ablation.

Hackathon demo. A live app on the edge device takes a webcam frame or uploaded clip and shows, side by side, the FP32 and INT8 verdicts, their attention heatmaps, and the measured latency—making the accuracy–efficiency–fidelity trade-off visible in seconds.

Tools. PyTorch, `timm`, `torchao`, ONNX Runtime, VS Code and GitHub, Google Colab GPU, Google Drive for data and checkpoints.

Pilot. Before full training, the team will verify dataset access, face extraction, one epoch of M0 training, and INT8 conversion of a single CBAM block. No pilot results exist yet.

Team responsibilities.

- **Kamar Bdair:** dataset access, face extraction, splits, data loaders.
- **Layan Ziyadah:** B1, M0, C1, M1 implementation and training.
- **Faisal Shamsi:** M2/M3 quantization, edge deployment, latency benchmarking.
- **Sedra Aljundi:** literature review, attention-fidelity and Grad-CAM analysis, evaluation scripts, documentation.
- **Dr. Naila Marir:** supervision, methodology review, and validation of the experimental protocol.

Timeline. *Week 1:* submit dataset requests on day 1; face extraction; pilot. *Week 2:* train B1, M0, C1, M1 (3 seeds). *Week 3:* PTQ (M2), edge benchmarking, M3 if on schedule. *Week 4:* fidelity analysis, cross-dataset tests, report. The plan will follow course deadlines.

8 Risks and Backup Plans

- **Dataset access delay:** FF++ and Celeb-DF require request forms, so both are submitted in week 1; DFDC preview is the fallback external set.
- **Limited GPU:** reduce frames per video and epochs equally for all models; keep three seeds for M0 and M1 at minimum.
- **CBAM not fully quantizable:** quantize supported layers, report the floating-point remainder, and treat this as evidence for M3.
- **No edge device:** report laptop-CPU latency and state the limitation clearly.
- **Overfitting to FF++ artifacts:** strong augmentation (compression, blur, color jitter) and cross-dataset testing.

Negative or null results (e.g., CBAM giving no gain) will be reported as findings, not hidden.

9 Expected Contributions

1. A controlled comparison separating the effect of spatial attention, channel attention and capacity in deepfake detection.
2. An *attention-fidelity* analysis showing whether INT8 quantization changes what a deepfake detector looks at.
3. An evaluation of a quantization-friendly CBAM variant.
4. Reproducible accuracy–efficiency results measured on real edge hardware, with code released on GitHub.
5. A live edge demo showing FP32 vs. INT8 verdicts and attention maps side by side.

10 AI-Use Statement

Generative AI assisted with literature understanding, experiment planning and draft organization. The team is responsible for verifying references, implementation, experimental validation and independent interpretation. No experimental results are presented as completed findings.

References

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